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Research on Micro-structure of Electroless Ni-B Coatings

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Abstract. By means of X-ray diffractometry (XRD) and transmission electron microscope (TEM), the micro-structure of 5 different Ni–B alloy coatings prepared by electroless plating with Potassium borohydride content ranging from 0.2 to approximately 1.4g/L was studied The influence of the KBH4 content in the coatings on their micro-structure was also investigated. It has been found that Ni–B alloy coatings showed amorphous sytuctue with XRD while with TEM it showed a combination of amorphous and nanocrystalline structure. It was further revealed that the amorphous structure increased with increasing KBH4 content, which means that Ni-B alloy coating prepared with increasing KBH4 content undergoes a transition process as follows: nanocrystalline → crystalline → amorphous structure. Results indicated that the micro-structure of Ni-B alloy coatings should be studied by both methods of XRD and TEM.

Introduction

Electroless plating is a chemical reduction process which depends upon the catalytic reduction process of metal ions in an aqueous solution (containing a chemical reducing agent) and the subsequent deposition of metal ions without the use of electrical energy. Due to the catalytic ability of metal coating, the chemical reduction process will keep working after the metal ions depositing on the surface of activated parts until the parts removing[1,2].

Electroless Ni–B alloy coating is extremely hard, incredible wear and abrasion resistance, which can be deposited on many kinds of Metallic and nonmetallic substrate. With the good performance in weldability, bonding, diffusion potential barrier effect, magnetic and electrical properties, Ni–B alloy coating can be used to substitute for gold-plated, silver, rhodium and Ge in electronic component, which will create considerate economic benefits by saving precious metal. More study should be carried out to promote the application of electroless plating Ni–B alloy coating. Some work has been done to investigate the plating process and performance of coatings by researchers, while few literature on micro-structure of Ni–B alloy coating was pubished, so it is quite significant to study the micro-structure of electroless plated Ni–B alloy coating[3,4].

Samples preparation

A polished 08F steel chip measured 50mm in length × 10mm in width × 0.2mm in height was used as the substrate of electroless plating.

Pretreatment

The procedure of pretreatment is as follows:

Sample polishing→cleaning in flowing cold water→degreasing in organic solvent→cleaning in flowing cold water→chemical degreasing→cleaning in flowing water of 40°C～60°C → cleaning in flowing cold water → electrolytic degreasing→ cleaning in flowing water of 40°C～60°C→cleaning in flowing cold water→weak corrosion→cleaning in flowing cold water → neutralizing → cleaning by distilled water.

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Immediately after pretreatment, sample was moved into electrolytic cell for electroless plating.

Tab.1 Solution component and process condition with five different KBH4 content.

Component \ Sample	1	2	3	4	5
KBH4(g/L)	0.2	0.5	0.9	1.1	1.4
NiCl(g/L)	30	30	30	30	30
C2H8N2(mL/L)	60	60	60	60	60
NaOH(g/L)	46	46	46	46	46
Temperature	85～90°C	85～90°C	85～90°C	85～90°C	85～90°C
PH value	13.5	13.5	13.5	13.5	13.5

Structure analysis of the coatings by XRD

Due to the low thickness of coatings,the X-ray may penetrate coatings to the substrate steel and add the Crystal structure information of steel on the coatings, resulting in the error of the X-ray analysis. When the angle θ formed by the crystal surface and the incident X-ray，the crystal surface spacing d，and the wavelength of ray λ conform to the Bragg equation $2d\sin\theta = n\lambda$, strong diffusion peak will appear and be detected by the instrument[5-7]. However, the equation does not define the relation between the angle that formed by the crystal surface and sample surface and incident X-ray, so angle α formed by incident X-ray and sample surface, azimuth ψ formed by crystal surface and sample surface are introduced, and $\psi = \theta - \alpha$. As shown in Fig 1. reflecting angle $\beta = 2\theta - \alpha$ When $\alpha = \beta \equiv \theta$, which meets the Bragg reflection principle, can lead to the θ-2θ symmetric coupled mode in the scanning process; yet $\alpha \neq \beta$ can lead to non-coupling mode by using asymmetric Bragg diffraction equipment. In the XRD analysis, both $\theta(\theta \mp \omega_0)/2\theta$ coupled mode and $\theta(\alpha_0)/2\theta$ non-coupling mode were used. In the coupled mode, $\psi \equiv \omega_0$, the depth penetrated by X-ray decreased with decreasing angle θ, yet in the non-coupled mode, let $\alpha = \alpha_0$, the depth penetrated by X-ray could be controlled by changing α_0, when $\sin\alpha_0 << \sin\beta$, $t = 0.13\sin\alpha_0/\mu$ (μ is the line absorption coefficient of the material), so the crystal structure information in different depth of sample could be obtained by adjusting the value of α_0.

The phase structure of the coatings were examined by X-ray diffraction analysis using SIEMENS X-ray diffractometer D5000 with Cu Kα1 target operated at 35KV and 40mA in the coupled mode, while in the non-coupled mode, the tube current and α_0 is 30mA and 2°, respectively.

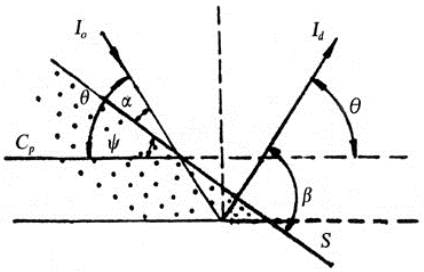


Fig.1 Asymmetric Prague diffraction pattern

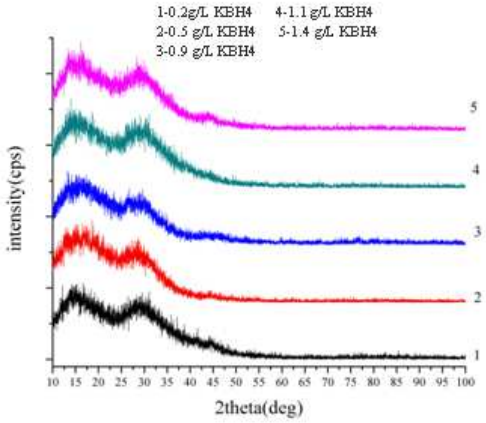


Fig.2 X-ray diffraction patterns of Ni–B coatings by using non-coupled mode

Fig.2 presents the X-ray diffraction patterns of Ni–B coatings with different KBH4 content by using non-coupled mode. Two main broadens peaks and very small peaks means that the Ni–B coatings with different KBH4 content are considered as amorphous.

Fig.3 presents the X-ray diffraction patterns of Ni–B coatings with different KBH4 content by using coupled mode, and for comparison, X-ray diffraction patterns of steel substrate is also shown in the figure. Because of the strong penetrating ability of X-ray, phase structure of steel substrate was shown in the results. Three distinct peaks which were identified as the diffraction peaks of Fe(110), Fe(200), Fe(211) arose, however, the peaks were seriously broadened, which is the typical characteristics of amorphous in the X-ray diffraction patterns.

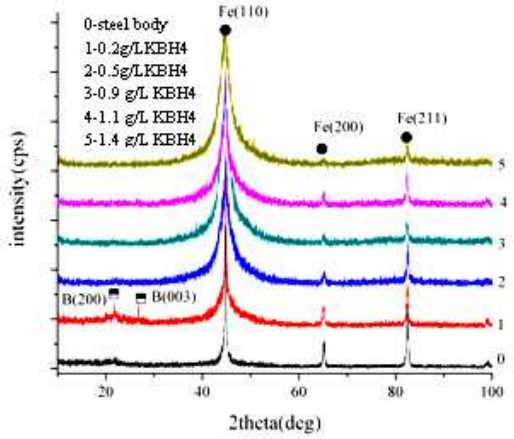


Fig.3 X-ray diffraction patterns of Ni–B coatingsby using coupled mode

Thus, amorphous phase is confirmed to exist in the Ni-B alloy coatings, however, it is still hard to find out if the structure of the samples are microcrystalline, amorphous or a mixture of microcrystalline and amorphous by using only XRD, some other method should be used.

Morphology analysis of the coatings by TEM

Transmission electron microscopy (TEM) can precisely observe the microstructure of samples and reveal the organized features through a comparative analysis of the electron diffraction image and electron microscopic image. The morphology of Ni-B alloy coatings were studied by transmission electron microscopy(JEOL JEM-3010 microscopy).

Fig.4 shows the TEM morphology and electron diffraction image of the Ni–B alloy coatings with 0.9g/L KBH4 content. Some clusters that consists of nanocrystal or several subgrain can be clearly observed, the average size of each cluster is 2-8nm, so it is called nanocrystalline.The electron diffraction image shows a lot of scattered diffraction rings, which means a lot of crystal existing in Ni-B alloy coatings. The brightest one of these diffraction rings is actually the superposition of Ni (200) and Ni (111) diffraction ring. On the other hand, there is amorphous phase but no big crystal in the coatings because the rings are not sharp. So, it can be seen that the Ni–B alloy coatings are actually a mixture of nanocrystalline and amorphous phase.

The microstructure of Ni-B alloy coatings is related with B content, with increasing B content, Ni-B alloy coating appeared to change from crystalline to amorphous phase[7,8], so the mixed phase is a transition state between crystalline to amorphous phase.

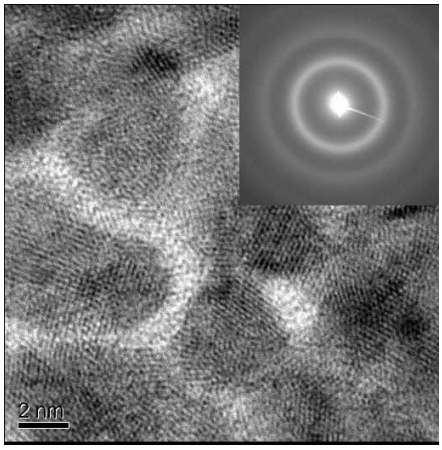


Fig.4 The TEM morphology and electron diffraction image of the Ni–B alloy coatings with 0.9g/L KBH4 content.

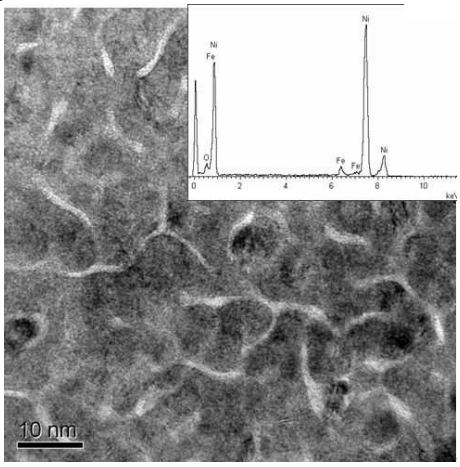


Fig.5 The mixed structure of Ni-B alloy coatings prepared with 0.9g/L KBH4

Fig.5 shows the mixed structure of Ni-B alloy coatings prepared with 0.9g/L KBH4. Incomplete grain boundaries can be clearly seen in this figure, which confirm that the Ni–B alloy coatings are actually a mixture of nanocrystalline and amorphous phase. In addition, energy spectrum analysis indicates that Ni element is the main component and little O element exists, so slight oxidation happens in this area. However, no B element is examined in the energy spectrum, because B is light element and hard to be examined by Energy Disperse Spectroscopy.

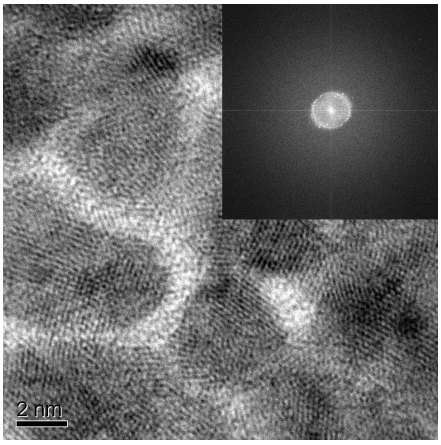


Fig. 6 The TEM morphology and electron diffraction image of the Ni–B alloy coatings with 0.9g/L KBH4 content after Fourier transform.

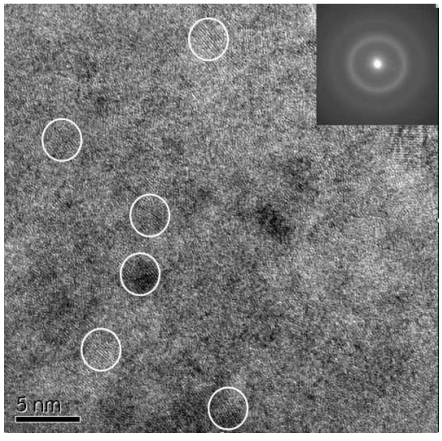


Fig.7 The TEM morphology and electron diffraction image of the Ni–B alloy coatings with 1.4g/L KBH4 content

Fig.6 shows the TEM morphology and electron diffraction image of the Ni–B alloy coatings with 0.9g/L KBH4 content after Fourier transform. Only one diffraction ring of Ni(111) can be seen, and the crystalline interplanar spacing is very close to the standard crystalline interplanar spacing of Ni(111), which means Ni crystal is the main component of the nanocluster, and no B element dissolves in Ni crystal, or the B element that is dissolved in Ni do not change the structure of Ni crystal.

Fig.7 shows the TEM morphology and electron diffraction image of the Ni–B alloy coatings with 1.4g/L KBH4 content. The coatings is mainly made of amorphous phase, nanocrystalline cluster is few in number. This is also confirmed by the electron diffraction image since the rings are almost the scattering diffraction rings. With the increase of B content, Ni-B alloy coating appeared to change from crystalline to amorphous phase, which is consistent with the analysis of Xuan TP.

Conclusions

(1) Amorphous phase exists in the electroless Ni-B alloy coatings by XRD analysis.

(2) TEM analysis indicates that the electroless Ni-B alloy coatings are a mixture of nanocrystalline and amorphous phase. with the increase of KBH4 content, nanocrystalline decreases while amorphous phase increases, Ni-B alloy coating appeared to go through a transition process: nanocrystalline→crystalline→ amorphous structure.

(3) It is hard to find out the accurate structure of electroless Ni-B alloy coatings only by XRD, no matter in coupled mode or non-coupled mode, TEM analysis should be done at the same time so as to get the correct answer.

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